

46-929, Fall 2019: Homework #2

Due 5:30 PM, Wednesday, November 6, 2019

Please look at the syllabus for the list of ways in which the HW should be turned in. **Also**, hand-written assignments are perfectly acceptable; of course, they should be clear.

Remember that you are to work **individually** on your homework assignments.

Include all relevant R commands in the homework that you hand in.

1. Do Exercise 2 on page 356 in Ruppert and Matteson. Note that Exercise 1 provides instructions for obtaining the data set.

When you answer part (a) for this question, be sure to include each of the following in your discussion:

- (a) An assessment of the normal probability plot of the residuals.
 - (b) An assessment of the ACF of the residuals.
 - (c) A comparison of the sample ACF for the data and the fit ACF for the model.
 - (d) Use of the Ljung-Box Test on the residuals.
 - (e) The value of AICc for each of the models.
2. For the previous question: Use `auto.arima()` to determine if there is an $AR(p)$ model which is better than either $p = 1$ or $p = 2$. Base your conclusion on AICc.

Remember to set `max.q = 0`.

Note: Use `approximation=F` to get your results. Also try using `approximation = T` to see if the results differ.

Problem 7 *Include the plot with your work.*

- (a) *Why do the prediction intervals (blue curves) widen as one moves farther into the future?*
- (b) *Why are the predictions (red) constant throughout?*

12.16 Exercises

1. This problem and the next use CRSP daily returns. First, get the data and plot the ACF in two ways:

```

1 library(Ecdat)
2 data(CRSPday)
3 crsp=CRSPday[,7]
4 acf(crsp)
5 acf(as.numeric(crsp))

```

- (a) Explain what “lag” means in the two ACF plots. Why does lag differ between the plots?
 - (b) At what values of lag are there significant autocorrelations in the CRSP returns? For which of these values do you think the statistical significance might be due to chance?
2. Next, fit AR(1) and AR(2) models to the CRSP returns:


```

6 arima(crsp,order=c(1,0,0))
7 arima(crsp,order=c(2,0,0))

```

 - (a) Would you prefer an AR(1) or an AR(2) model for this time series? Explain your answer.
 - (b) Find a 95% confidence interval for ϕ for the AR(1) model.
 3. Consider the AR(1) model

$$Y_t = 5 - 0.55Y_{t-1} + \epsilon_t$$

and assume that $\sigma_\epsilon^2 = 1.2$.

- (a) Is this process stationary? Why or why not?
 - (b) What is the mean of this process?
 - (c) What is the variance of this process?
 - (d) What is the covariance function of this process?
4. Suppose that $Y_1, Y_2, \dots$ is an AR(1) process with $\mu = 0.5$, $\phi = 0.4$, and $\sigma_\epsilon^2 = 1.2$.
 - (a) What is the variance of Y_1 ?
 - (b) What are the covariances between Y_1 and Y_2 and between Y_1 and Y_3 ?
 - (c) What is the variance of $(Y_1 + Y_2 + Y_3)/2$?
 5. An AR(3) model has been fit to a time series. The estimates are $\hat{\mu} = 104$, $\hat{\phi}_1 = 0.4$, $\hat{\phi}_2 = 0.25$, and $\hat{\phi}_3 = 0.1$. The last four observations were $Y_{n-3} = 105$, $Y_{n-2} = 102$, $Y_{n-1} = 103$, and $Y_n = 99$. Forecast Y_{n+1} and Y_{n+2} using these data and estimates.